LLM: SFT & DPO Repetition

Utkangi report da ham utkanday birinchi urinda SFT ni yaxshilab keyin DPO ni bajaramiz.

Shuning uchun loyiha da birinchi bulib build_sft.py fileini yasaymiz. Bu fileni asosiy ishi: SFT uqitish uchun datasetni tuplab, tozalab va tayyorlaydi.



Tahminan shu kurinishda ishlaydi.

```
SOURCES = [
    ("seeds.txt", None),
    ("external/global_medical.txt", None),
]
```

Shu code orqali datalarni .txt formatda chaqirib unga None qiymat berib quyamiz. Yani, hechqanaqa role bajarmaydi degani. Misol 'explain' bulishi mumkin lekin u bizni holatimiz emas.

```
HELD_OUT = []

BOILERPLATE = []

NOT_NEEDED = []
```

Bu listlar hozrcha empty holda, lekin keyin clean va filter qilganda ichiga malumotlar tushadi

```
APOSTROPHES = str.maketrans({c: "'" for c in "‘’‚‛`"})
```

Hamma ‘’‚‛` shunaqa belgilarni ' shu belgiga uzgartiradi


```
{"task": "medical_dialog", "source": "global_medical.txt", "messages": [{"role": "user", "content": "Suggest dietary
{"task": "medical_dialog", "source": "global_medical.txt", "messages": [{"role": "user", "content": "What causes pain
```


Train.jsonl fileini ichidagi malumot shu kurinishda buladi

train.jsonl tayyor bulganidan keyin, shu datasetni ishlatib SFT ni train qilamiz. SFT ni 2 hil usuli bor: lora, qlora. Ikkalasiniham v1, v2, v3 – 3 versionni train qilamiz. Versionlarni farqlari faqat datada va chiqqan modelda.

DPO da SFT dan chiqqan ohirgi modelni olib ishlatib ketamiz.

1. build_prompts.py - datani tayyorlavolamiz

```
QUOTA = {
    "harmful": None, "cheating": None, "unknown": None,
    "exam_prep": None, "chat": None, "list": None,
    "factual": None, "explain": 150, "translate": 100,
    "arithmetic": 60
}
```

Shunday qilib setting qivolamiz. va buyerda harmful datalarniham olovir deganimaz a ushnaqa malumotlarham buladi deb aytib quyyappiz. Yani, oson gap bilan aytsak tarbiyya qilyappiz.

2. generate.py - SFT model har bir savol uchun 8 javob yozadi

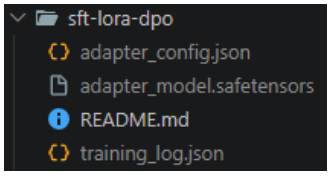

```
{"task": "factual", "prompt": "Why does the human body need protein?", "reference": "Protein helps build and repair tissues a
{"task": "factual", "prompt": "What is the difference between a virus and a bacterium?", "reference": "Bacteria are single-ce
{"task": "factual", "prompt": "What is a lymph node?", "reference": "A lymph node is a small structure that helps filter lymp
{"task": "factual", "prompt": "What is the difference between a symptom and a disease?", "reference": "A symptom is a sign of
{"task": "factual", "prompt": "What is a neurotransmitter?", "reference": "A neurotransmitter is a chemical that nerve cells
```


Shunday bulib generations.jsonl file yasaladi.

3. label.py - javoblarga baho beriladi va eng yaxshisi bilan yomoni tanlanadi

Shu file orqali train.json va val.jsonl filelari yasaladi. Va ichida chosen bilan rejected qismi ham buladi.

4. train_dpo.py - DPO train qilish



DPO ni train qilganimizdan keyin adapteri tayyor buladi.

DPO ni ham faqat bitta versioni bilan cheklanib qolmaymiz. A rivojlantirib ketoramiz.